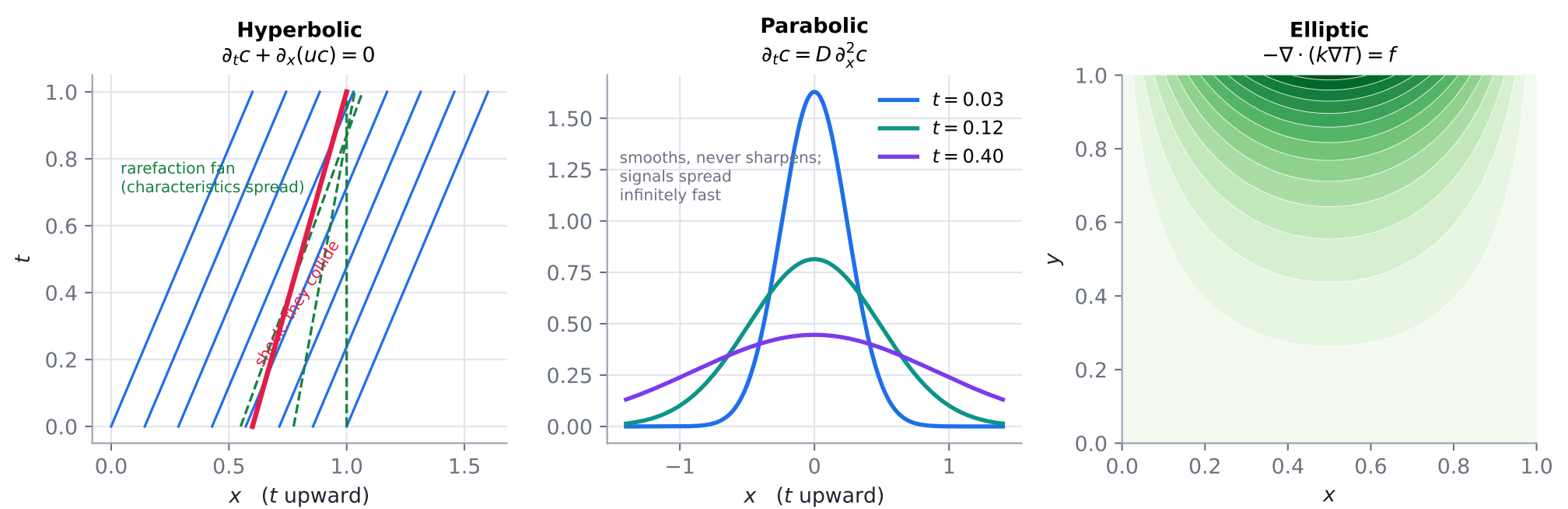

Hyperbolic
$\partial_t c + \partial_x(uc) = 0$
rarefaction fan
(characteristics spread)
shock: they collide
t
x (t upward)
Parabolic
$\partial_t c = D\,\partial_x^2 c$
smooths, never sharpens;
signals spread
infinitely fast
$t = 0.03$
$t = 0.12$
$t = 0.40$
x (t upward)
Elliptic
$-\nabla \cdot (k\nabla T) = f$
y
x